

Independent Vector Analysis (IVA): Concepts, Applications, and Computation of Demixing Matrices

Abstract

Independent Vector Analysis (IVA) extends Independent Component Analysis (ICA) to handle multiple datasets simultaneously by leveraging dependencies between them. This document explores the fundamental concepts of IVA, the nature of datasets involved, mutual information computations, and the process of computing demixing matrices through gradient-based optimization. An illustrative example is provided to clarify the independence of sources within each dataset and their dependencies across datasets.

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1 Introduction to Independent Vector Analysis (IVA)

Independent Vector Analysis (IVA) is a computational method used to separate mixed signals into their underlying independent sources across multiple datasets. Unlike Independent Component Analysis (ICA), which operates on a single dataset, IVA simultaneously analyzes multiple datasets, exploiting the dependencies between them to enhance source separation quality.

1.1 Relationship Between IVA and ICA

- **ICA:** Focuses on separating independent components within a single dataset, assuming that the observed data are linear mixtures of unknown independent sources.
- **IVA:** Extends ICA to multiple datasets, allowing for the exploitation of dependencies between datasets to achieve more accurate and robust source separation.

Mathematically, IVA is represented as:

$$\mathbf{X}_k = \mathbf{A}_k \mathbf{S}_k \quad \text{for } k = 1, 2, \dots, K$$

where:

- $\mathbf{X}_k$ is the observed data matrix for dataset k .
- $\mathbf{A}_k$ is the unknown mixing matrix for dataset k .
- $\mathbf{S}_k$ contains the independent source signals for dataset k .

2 Nature of Datasets in IVA

In IVA, each dataset $\mathbf{X}_k$ represents a different mixture or representation of the same underlying independent sources. This setup allows IVA to leverage the shared information across datasets for more effective source separation.

2.1 Practical Examples of Datasets $\mathbf{X}_k$

- **Audio Signal Processing:**
 - **Scenario:** Blind Source Separation (BSS) of audio signals recorded in multiple environments or conditions.
 - **Datasets $\mathbf{X}_k$:** Each $\mathbf{X}_k$ corresponds to recordings from the same set of microphones in different acoustic environments (e.g., different rooms, different noise conditions, or different time sessions).
 - **Underlying Sources:** The same set of speakers or sound sources appearing across all environments, but with different mixing characteristics in each environment.
- **Biomedical Signal Processing:**
 - **Scenario:** Joint analysis of EEG and MEG data.

- **Datasets $\mathbf{X}_k$:** One dataset might consist of EEG recordings, another of MEG recordings.
- **Underlying Sources:** Neural activity patterns, brain rhythms, etc.
- **Multimodal Data Integration:**
 - **Scenario:** Combining audio and visual data for tasks like video enhancement.
 - **Datasets $\mathbf{X}_k$:** One dataset could be the audio track, another the visual frames.
 - **Underlying Sources:** Simultaneous audio and visual events.
- **Remote Sensing:**
 - **Scenario:** Integrating data from various sensors (e.g., optical, infrared, radar).
 - **Datasets $\mathbf{X}_k$:** Each dataset represents data from a different sensor.
 - **Underlying Sources:** Different features of the landscape, vegetation types, water bodies, etc.
- **Financial Data Analysis:**
 - **Scenario:** Integrating multiple financial indicators for market prediction.
 - **Datasets $\mathbf{X}_k$:** Different datasets might represent stock prices, trading volumes, and economic indicators.
 - **Underlying Sources:** Market trends, investor behaviors, macroeconomic factors.

2.2 Consistency of Sources Across Datasets

- **Consistent Sources:** All datasets share the same set of underlying sources, mixed differently. This one-to-one correspondence facilitates the exploitation of dependencies between corresponding sources.
- **Varying Sources:** Different datasets may capture different subsets of sources or represent sources in varying forms. Extended IVA models may handle such cases by accommodating varying source counts or partial correspondences.

3 Independence Assumptions in IVA

In IVA, the independence assumptions are more nuanced compared to ICA due to the presence of multiple datasets:

3.1 Within Each Dataset

Sources Are Independent: Similar to ICA, within each individual dataset $\mathbf{X}_k$, the sources $\mathbf{S}_k$ are assumed to be statistically independent. This means that for a single dataset, the sources do not share any statistical dependencies among themselves.

3.2 Across Datasets

- **Sources Are Dependent Across Datasets:** For a given source, its representation across different datasets is dependent. This implies that the same underlying source manifests in each dataset, preserving some form of dependency or consistency. For example, the sound of a particular instrument captured by different microphones would have consistent characteristics across recordings.
- **Different Sources Are Independent Across Datasets:** Different sources remain independent from each other across all datasets. This means that while corresponding sources across datasets are dependent, distinct sources do not influence each other, maintaining overall independence in the system.

3.3 Illustrative Example

Consider two datasets $\mathbf{X}_1$ and $\mathbf{X}_2$, each containing two independent sources.

3.3.1 Dataset 1

$$\begin{aligned}\mathbf{X}_1 &= \mathbf{A}_1 \mathbf{S}_1 \\ \mathbf{S}_1 &= [s_{11} \quad s_{12}]^T \quad (\text{Two independent sources})\end{aligned}$$

3.3.2 Dataset 2

$$\begin{aligned}\mathbf{X}_2 &= \mathbf{A}_2 \mathbf{S}_2 \\ \mathbf{S}_2 &= [s_{21} \quad s_{22}]^T \quad (\text{Two independent sources})\end{aligned}$$

3.3.3 IVA Assumptions

- **Within Each Dataset:**

$$s_{11} \perp s_{12}, \quad s_{21} \perp s_{22}$$

This denotes that within each dataset, the sources are independent.

- **Across Datasets:**

$$s_{11} \not\perp s_{21}, \quad s_{12} \not\perp s_{22}$$

However,

$$s_{11} \perp s_{22}, \quad s_{12} \perp s_{21}$$

This structured dependency allows IVA to effectively separate sources by leveraging the consistent information across multiple datasets.

4 Mutual Information

Mutual Information (MI) quantifies the amount of information one random variable contains about another, capturing both linear and non-linear dependencies. In IVA, MI is used to measure and minimize dependencies between estimated sources across datasets to achieve statistical independence.

4.1 Mutual Information Between Two Vectors

For two random vectors $\mathbf{X} \in \mathbb{R}^n$ and $\mathbf{Y} \in \mathbb{R}^m$, the mutual information $I(\mathbf{X}; \mathbf{Y})$ is defined as:

$$I(\mathbf{X}; \mathbf{Y}) = \int_{\mathcal{X}} \int_{\mathcal{Y}} p(\mathbf{x}, \mathbf{y}) \log \left(\frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})} \right) d\mathbf{x}d\mathbf{y}$$

where:

- $p(\mathbf{x}, \mathbf{y})$ is the joint probability density function (PDF) of $\mathbf{X}$ and $\mathbf{Y}$.
- $p(\mathbf{x})$ and $p(\mathbf{y})$ are the marginal PDFs of $\mathbf{X}$ and $\mathbf{Y}$, respectively.

4.2 Mutual Information Between Two Matrices

For two matrices $\mathbf{S}_1 \in \mathbb{R}^{n_1 \times m_1}$ and $\mathbf{S}_2 \in \mathbb{R}^{n_2 \times m_2}$, mutual information can be computed in various ways:

- **Row-wise MI:** Considering each row as a separate sample, compute MI between corresponding rows across matrices.
- **Column-wise MI:** Considering each column as a separate feature, compute MI between corresponding columns across matrices.
- **Overall MI:** Treating entire matrices as high-dimensional random variables.

In the context of IVA, row-wise MI is typically more meaningful as rows often represent samples.

5 Solving the IVA Problem Using Mutual Information

The objective in IVA is to find demixing matrices $\mathbf{W}_k$ that minimize the mutual information between the estimated sources $\hat{\mathbf{S}}_k$, ensuring their statistical independence across datasets.

5.1 Objective Function

The cost function to be minimized is:

$$J(\{\mathbf{W}_k\}) = I(\hat{\mathbf{S}}_1; \hat{\mathbf{S}}_2; \dots; \hat{\mathbf{S}}_K)$$

where:

$$\hat{\mathbf{S}}_k = \mathbf{W}_k \mathbf{X}_k \quad \text{for } k = 1, 2, \dots, K$$

5.2 Gradient Descent Optimization

To minimize J , gradient descent can be employed. The update rule for each demixing matrix $\mathbf{W}_k$ is:

$$\mathbf{W}_k^{(t+1)} = \mathbf{W}_k^{(t)} - \eta \frac{\partial J}{\partial \mathbf{W}_k}$$

where η is the learning rate.

5.2.1 Computing Gradients

Assuming differentiable joint distributions, the gradient of J with respect to $\mathbf{W}_k$ is:

$$\frac{\partial J}{\partial \mathbf{W}_k} = \mathbf{W}_k^{-T} - \mathbb{E} \left[\hat{\mathbf{S}}_k \mathbf{X}_k^T \right]$$

where $\mathbf{W}_k^{-T}$ is the inverse transpose of $\mathbf{W}_k$.

5.2.2 Update Rules

The demixing matrices are updated iteratively as follows:

$$\mathbf{W}_k^{(t+1)} = \mathbf{W}_k^{(t)} - \eta \frac{\partial J}{\partial \mathbf{W}_k}$$

6 Illustrative Example: Computing Demixing Matrices

To clarify the independence of sources within each dataset and their dependencies across datasets, consider the following example.

6.1 Example Setup

Assume two datasets $\mathbf{X}_1 \in \mathbb{R}^{2 \times 3}$ and $\mathbf{X}_2 \in \mathbb{R}^{4 \times 5}$, each containing two independent sources.

6.1.1 Given Data

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2 \times 3}$$

$$\mathbf{X}_2 = \begin{bmatrix} 7 & 8 & 9 & 10 & 11 \\ 12 & 13 & 14 & 15 & 16 \\ 17 & 18 & 19 & 20 & 21 \\ 22 & 23 & 24 & 25 & 26 \end{bmatrix} \in \mathbb{R}^{4 \times 5}$$

Assume each dataset has two independent sources.

6.1.2 Assumptions

- **Within Each Dataset:** The sources within each dataset are independent.
- **Across Datasets:** Corresponding sources across datasets are dependent, while different sources remain independent.

6.2 Step-by-Step Computation

6.2.1 Step 1: Initialize Demixing Matrices

$$\mathbf{W}_1^{(0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{W}_2^{(0)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

6.2.2 Step 2: Compute Estimated Sources

$$\hat{\mathbf{S}}_1^{(0)} = \mathbf{W}_1^{(0)} \mathbf{X}_1 = \mathbf{X}_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\hat{\mathbf{S}}_2^{(0)} = \mathbf{W}_2^{(0)} \mathbf{X}_2 = \begin{bmatrix} 7 & 8 & 9 & 10 & 11 \\ 12 & 13 & 14 & 15 & 16 \end{bmatrix} \in \mathbb{R}^{2 \times 5}$$

6.2.3 Step 3: Compute Mutual Information

To compute the mutual information between $\hat{\mathbf{S}}_1^{(0)}$ and $\hat{\mathbf{S}}_2^{(0)}$, consider each row as a vector.

6.2.4 Assumptions for Simplicity

- Gaussian distribution of sources.
- Zero cross-covariance implies independence.

6.2.5 Compute Covariance Matrices

$$\Sigma_{\hat{\mathbf{S}}_1^{(0)}} = \frac{1}{3} \mathbf{X}_1 \mathbf{X}_1^T = \frac{1}{3} \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} = \begin{bmatrix} 4.6667 & 10.6667 \\ 10.6667 & 25.6667 \end{bmatrix}$$

$$\Sigma_{\hat{\mathbf{S}}_2^{(0)}} = \frac{1}{5} \mathbf{X}_2 \mathbf{X}_2^T = \frac{1}{5} \begin{bmatrix} 435 & 495 & 555 & 615 \\ 495 & 565 & 635 & 705 \\ 555 & 635 & 715 & 795 \\ 615 & 705 & 795 & 885 \end{bmatrix} = \begin{bmatrix} 87 & 99 & 111 & 123 \\ 99 & 113 & 127 & 141 \\ 111 & 127 & 143 & 159 \\ 123 & 141 & 159 & 177 \end{bmatrix}$$

$$\Sigma_{\hat{\mathbf{S}}_1^{(0)}, \hat{\mathbf{S}}_2^{(0)}} = \mathbf{0} \quad (\text{Assuming zero cross-covariance for independence})$$

6.2.6 Compute Mutual Information for Each Corresponding Pair

$$I(\hat{\mathbf{S}}_1^{(0)}; \hat{\mathbf{S}}_2^{(0)}) = \frac{1}{2} \log \left(\frac{|\Sigma_{\hat{\mathbf{S}}_1^{(0)}}| |\Sigma_{\hat{\mathbf{S}}_2^{(0)}}|}{|\Sigma_{\hat{\mathbf{S}}_1^{(0)}, \hat{\mathbf{S}}_2^{(0)}}|} \right)$$

Given zero cross-covariance:

$$|\Sigma_{\hat{\mathbf{S}}_1^{(0)}, \hat{\mathbf{S}}_2^{(0)}}| = |\Sigma_{\hat{\mathbf{S}}_1^{(0)}}| \cdot |\Sigma_{\hat{\mathbf{S}}_2^{(0)}}|$$

Thus:

$$I(\hat{\mathbf{S}}_1^{(0)}; \hat{\mathbf{S}}_2^{(0)}) = \frac{1}{2} \log(1) = 0 \text{ bits}$$

Conclusion: Assuming zero cross-covariance implies zero mutual information, indicating independence between datasets. However, in practice, dependencies are present, necessitating the estimation of cross-covariance from data.

6.3 Step 4: Compute Gradients

$$\frac{\partial J}{\partial \mathbf{W}_1} = \mathbf{W}_1^{-T} - \mathbb{E} [\hat{\mathbf{S}}_1^{(0)} \mathbf{X}_1^T]$$

Assuming $\mathbf{W}_1^{(0)} = \mathbf{I}$:

$$\mathbf{W}_1^{-T} = \mathbf{I}$$

$$\mathbb{E} [\hat{\mathbf{S}}_1^{(0)} \mathbf{X}_1^T] = \frac{1}{3} \mathbf{X}_1 \mathbf{X}_1^T = \frac{1}{3} \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} = \begin{bmatrix} 4.6667 & 10.6667 \\ 10.6667 & 25.6667 \end{bmatrix}$$

Thus:

$$\frac{\partial J}{\partial \mathbf{W}_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 4.6667 & 10.6667 \\ 10.6667 & 25.6667 \end{bmatrix} = \begin{bmatrix} -3.6667 & -10.6667 \\ -10.6667 & -24.6667 \end{bmatrix}$$

6.4 Step 5: Update Demixing Matrix $\mathbf{W}_1$

Using gradient descent with learning rate $\eta = 0.1$:

$$\mathbf{W}_1^{(1)} = \mathbf{W}_1^{(0)} - \eta \cdot \frac{\partial J}{\partial \mathbf{W}_1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} -3.6667 & -10.6667 \\ -10.6667 & -24.6667 \end{bmatrix} = \begin{bmatrix} 1 + 0.3667 & 0 + 1.0667 \\ 0 + 1.0667 & 1 + 2.4667 \end{bmatrix} = \begin{bmatrix} 1.3667 & 1.0667 \\ 1.0667 & 3.4667 \end{bmatrix}$$

6.5 Step 6: Iterate Until Convergence

Repeat Steps 2 to 5 iteratively, updating $\mathbf{W}_1$ and $\mathbf{W}_2$ until mutual information stabilizes, indicating that the sources have been sufficiently separated.

7 Mutual Information Computations for Simple Gaussian Examples

To elucidate the concept of mutual information, we compute it for two simple examples involving Gaussian distributions.

7.1 Example 1: Independent Gaussian Variables

Consider two independent Gaussian random variables:

$$X \sim \mathcal{N}(0, 1), \quad Y \sim \mathcal{N}(1, 3)$$

Since X and Y are independent, their joint distribution is the product of their marginal distributions:

$$p(X, Y) = p(X)p(Y)$$

$$p(X) = \frac{1}{\sqrt{2\pi}} e^{-\frac{X^2}{2}}, \quad p(Y) = \frac{1}{\sqrt{6\pi}} e^{-\frac{(Y-1)^2}{6}}$$

Thus, the mutual information $I(X; Y)$ is:

$$I(X; Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(X, Y) \log \left(\frac{p(X, Y)}{p(X)p(Y)} \right) dX dY = 0$$

Conclusion: The mutual information between two independent Gaussian variables is zero, indicating no dependency.

7.2 Example 2: Dependent Gaussian Variables

Now, consider two dependent Gaussian random variables:

$$X \sim \mathcal{N}(0, 1), \quad Y \sim \mathcal{N}(1, 3)$$

with covariance $\text{Cov}(X, Y) = \rho\sqrt{1 \times 3} = \rho\sqrt{3}$, where ρ is the correlation coefficient.

7.2.1 Joint Distribution

The joint distribution of X and Y is given by:

$$p(X, Y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{(X-\mu_X)^2}{\sigma_X^2} - 2\rho\frac{(X-\mu_X)(Y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(Y-\mu_Y)^2}{\sigma_Y^2}\right)\right)$$

where:

$$\mu_X = 0, \quad \sigma_X^2 = 1, \quad \mu_Y = 1, \quad \sigma_Y^2 = 3$$

7.2.2 Mutual Information Formula for Gaussian Variables

For two jointly Gaussian random variables, the mutual information $I(X; Y)$ is given by:

$$I(X; Y) = -\frac{1}{2} \log(1 - \rho^2)$$

where ρ is the correlation coefficient between X and Y .

7.2.3 Computing MI with Specific Correlation

Let's compute $I(X; Y)$ for $\rho = 0.5$:

$$I(X; Y) = -\frac{1}{2} \log(1 - 0.5^2) = -\frac{1}{2} \log(1 - 0.25) = -\frac{1}{2} \log(0.75)$$

$$\log(0.75) \approx -0.28768 \quad (\text{using natural logarithm})$$

$$I(X; Y) = -\frac{1}{2} \times (-0.28768) = 0.14384 \text{ nats}$$

To convert nats to bits, recall that $1 \text{ nat} \approx 1.4427 \text{ bits}$:

$$I(X; Y) \approx 0.14384 \times 1.4427 \approx 0.207 \text{ bits}$$

Conclusion: With a correlation coefficient of 0.5, the mutual information between X and Y is approximately 0.207 bits, indicating a moderate dependency.

8 Cost Function for IVA

The cost function for IVA can be written as the negative log-likelihood of the joint density minus terms that ensure the invertibility of the demixing matrices:

$$J(\{\mathbf{W}_k\}) = -\sum_{k=1}^K \mathbb{E} \left[\log p(\hat{\mathbf{S}}_k) \right] + \sum_{k=1}^K \log |\det(\mathbf{W}_k)|$$

where:

- $p(\hat{\mathbf{S}}_k)$ is the density of the estimated sources for dataset k .
- $\mathbf{W}_k$ is the demixing matrix for dataset k .

This formulation aims to maximize the likelihood of the independent sources while maintaining the invertibility of the demixing matrices.

9 Computing Demixing Matrices

The objective in IVA is to find demixing matrices $\mathbf{W}_k$ that minimize the mutual information between the estimated sources $\hat{\mathbf{S}}_k$, ensuring their statistical independence across datasets.

9.1 Gradient Descent Optimization

To minimize the cost function $J(\{\mathbf{W}_k\})$, gradient descent can be employed. The update rule for each demixing matrix $\mathbf{W}_k$ is:

$$\mathbf{W}_k^{(t+1)} = \mathbf{W}_k^{(t)} - \eta \frac{\partial J}{\partial \mathbf{W}_k}$$

where η is the learning rate.

9.1.1 Computing Gradients

Assuming differentiable joint distributions, the gradient of J with respect to $\mathbf{W}_k$ is:

$$\frac{\partial J}{\partial \mathbf{W}_k} = \mathbf{W}_k^{-T} - \mathbb{E} [\hat{\mathbf{S}}_k \mathbf{X}_k^T]$$

where $\mathbf{W}_k^{-T}$ is the inverse transpose of $\mathbf{W}_k$.

9.1.2 Update Rules

The demixing matrices are updated iteratively as follows:

$$\mathbf{W}_k^{(t+1)} = \mathbf{W}_k^{(t)} - \eta \frac{\partial J}{\partial \mathbf{W}_k}$$

10 Conclusion

Independent Vector Analysis (IVA) provides a robust framework for separating independent sources from multiple datasets by minimizing the mutual information between estimated sources. By leveraging the dependencies between datasets, IVA can achieve better separation quality than traditional ICA applied independently to each dataset. The document provided an overview of IVA, discussed the nature of datasets used in practice, illustrated mutual information computations for simple Gaussian examples, and detailed the initial steps of the gradient descent optimization process for computing demixing matrices.

11 References

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